

Gastric Cancer Multi-Omics

Complete figure set — every image standalone, one per page

29 figures · generated 02 August 2026
each reproduced from its committed source file

Figure 1

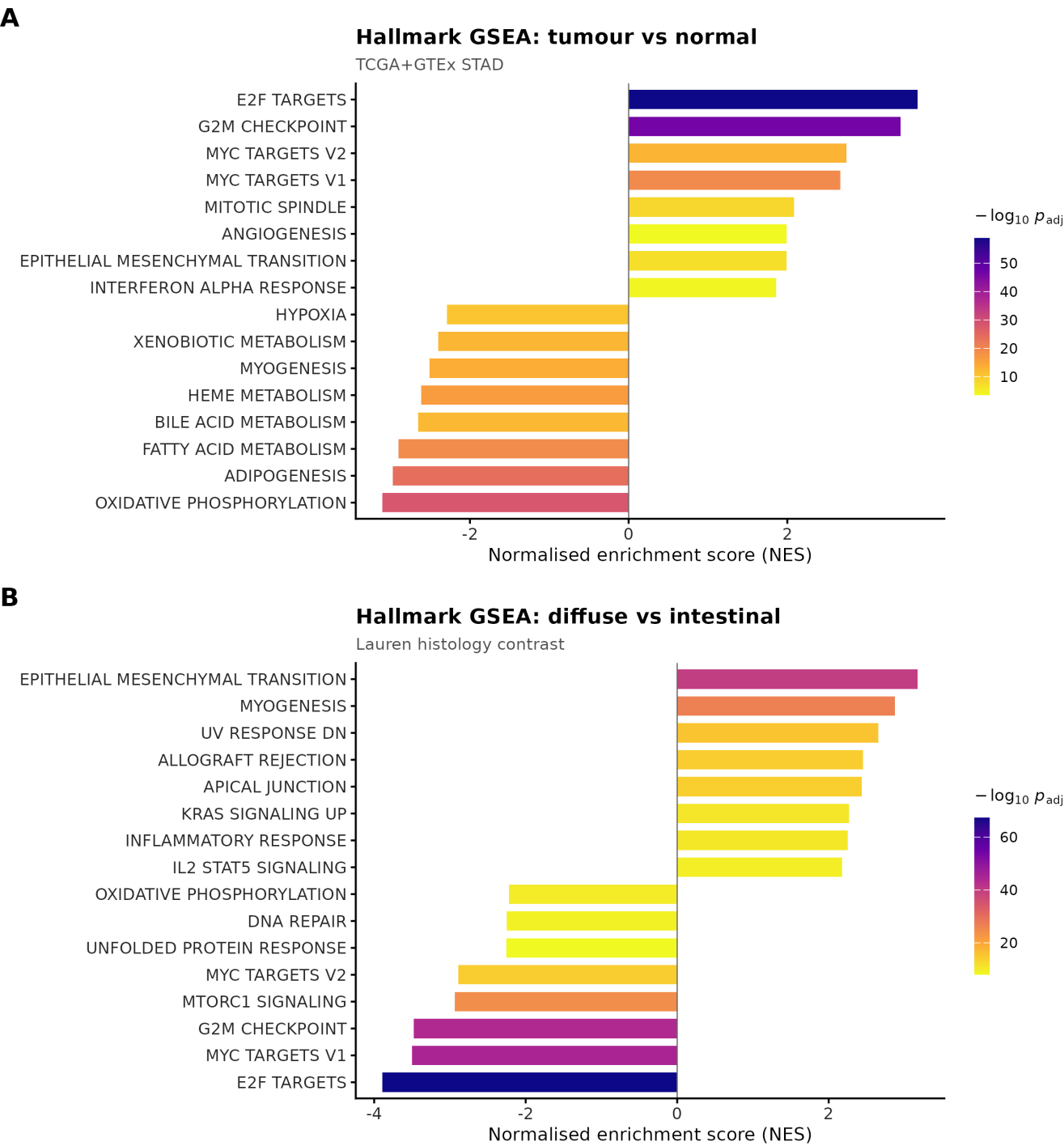


Figure 1. Tumour transcriptional programmes.

results/figures/Fig1.png

Figure 2

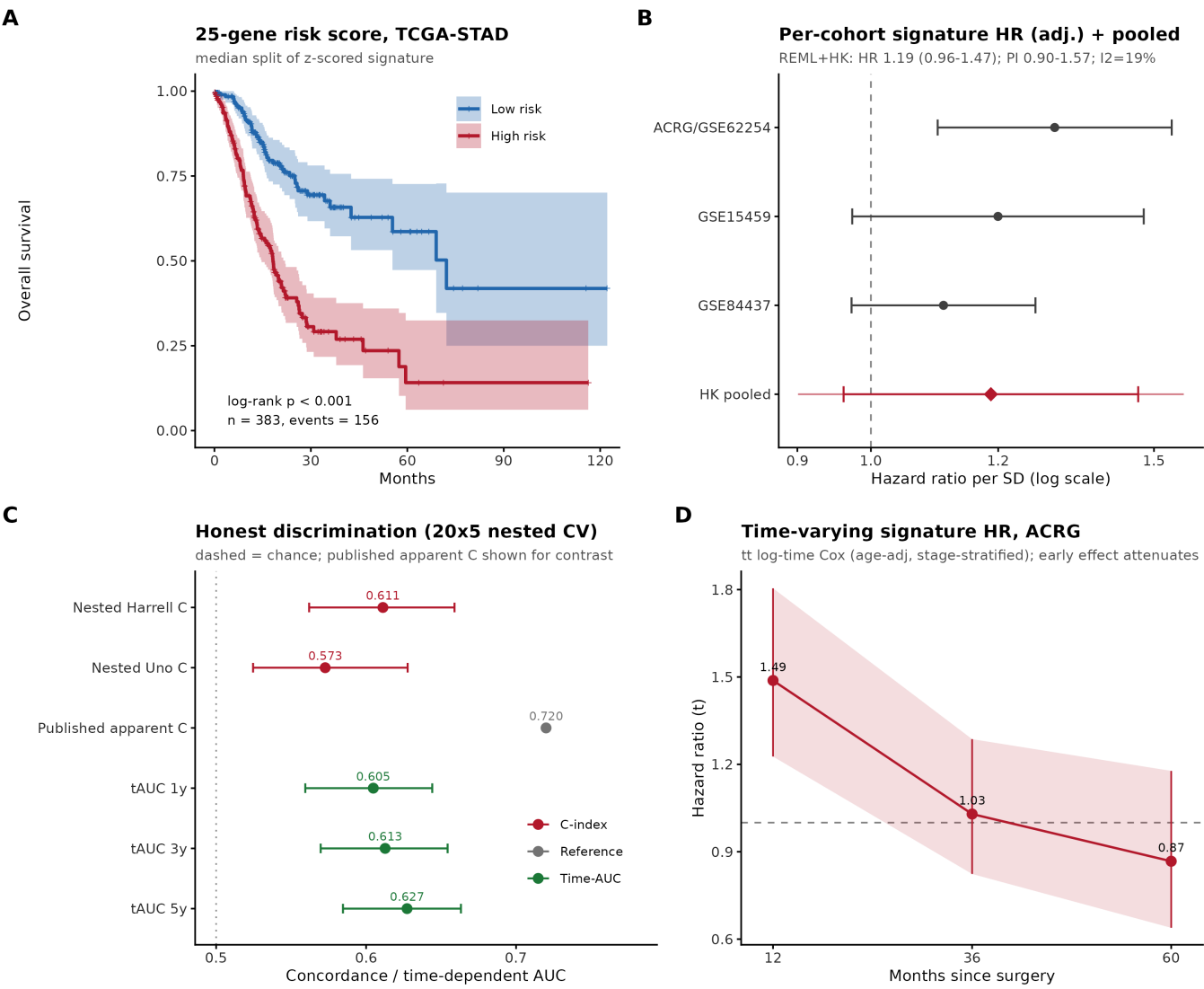


Figure 2. The 25-gene signature, reported honestly.
results/figures/Fig2.png

Figure 3

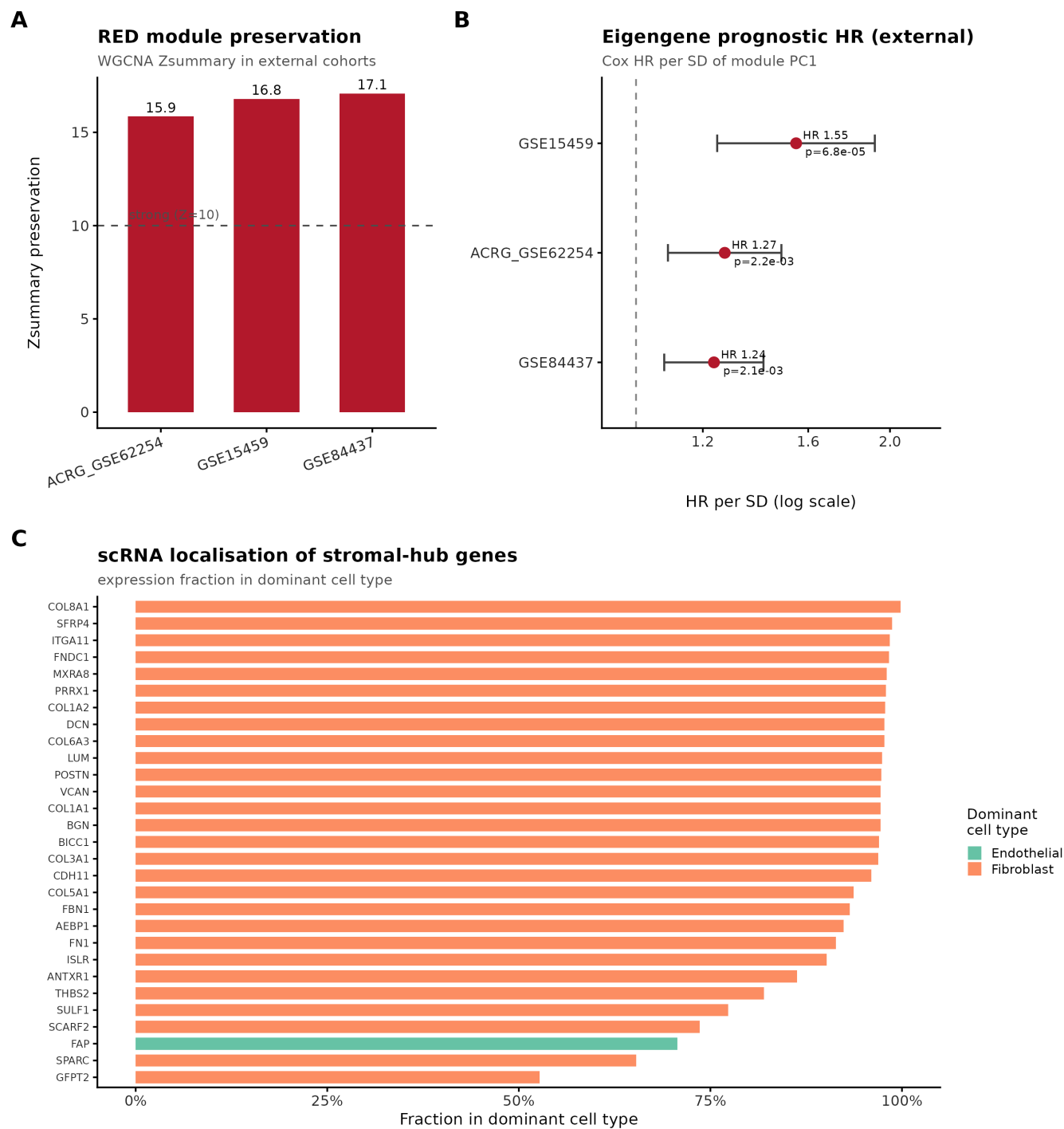


Figure 3. The externally-validated fibroblast/stromal module (primary finding).

results/figures/Fig3.png

Figure 4

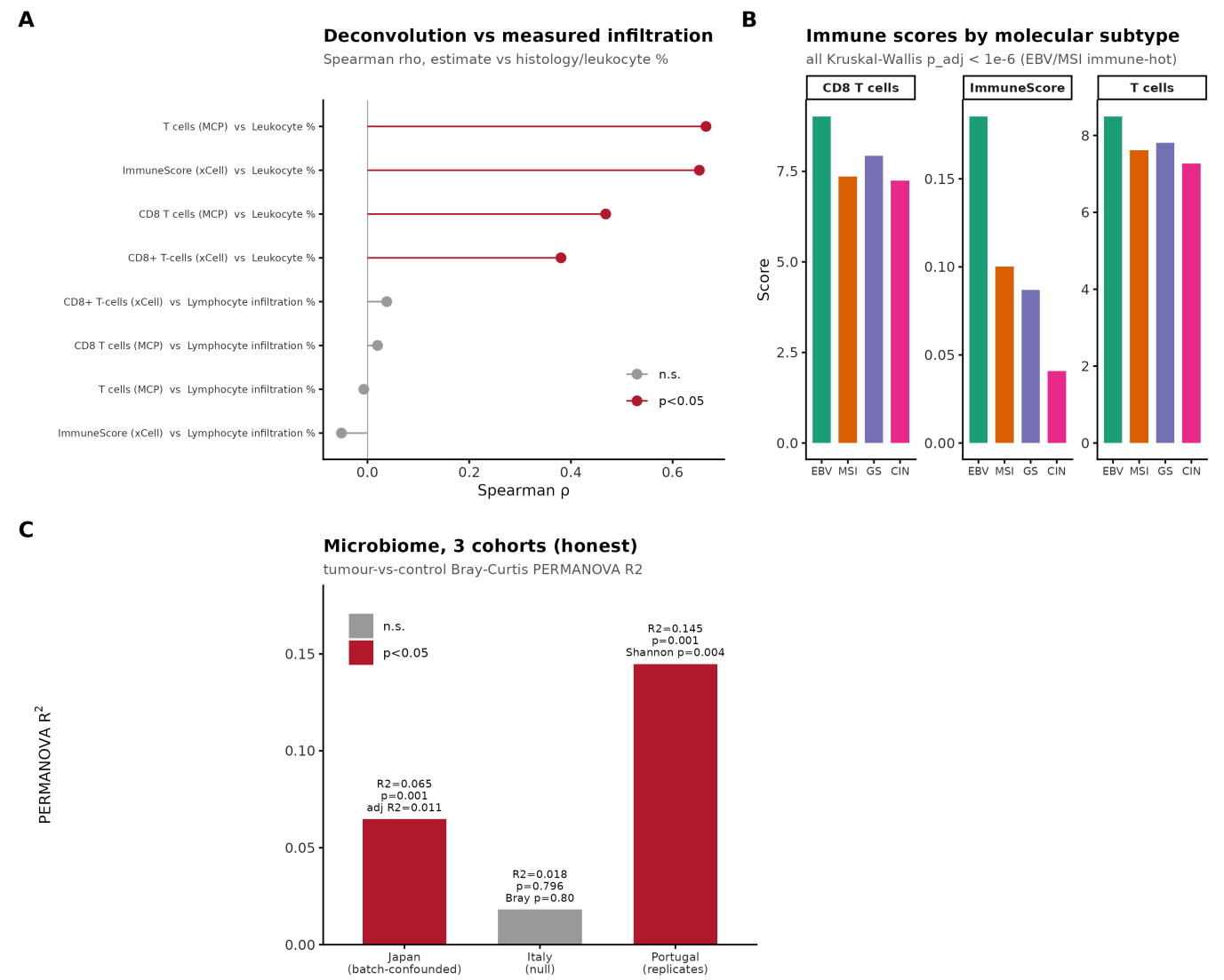


Figure 4. Immune microenvironment and the multi-cohort microbiome.
results/figures/Fig4.png

Figure 5

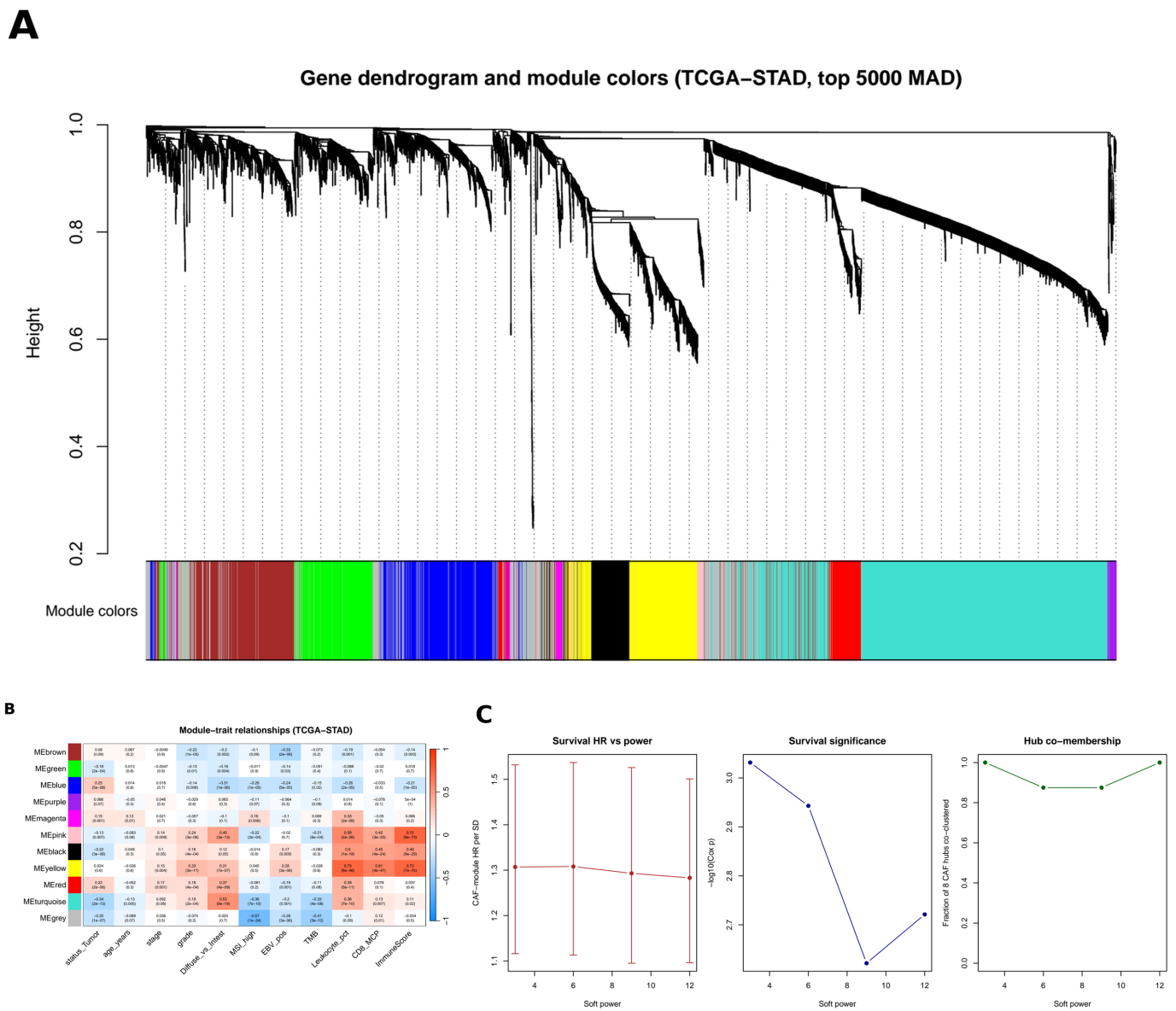


Figure 5. Weighted gene co-expression network analysis (WGCNA) of TCGA-STAD.

results/figures/fig5.png

Figure 6

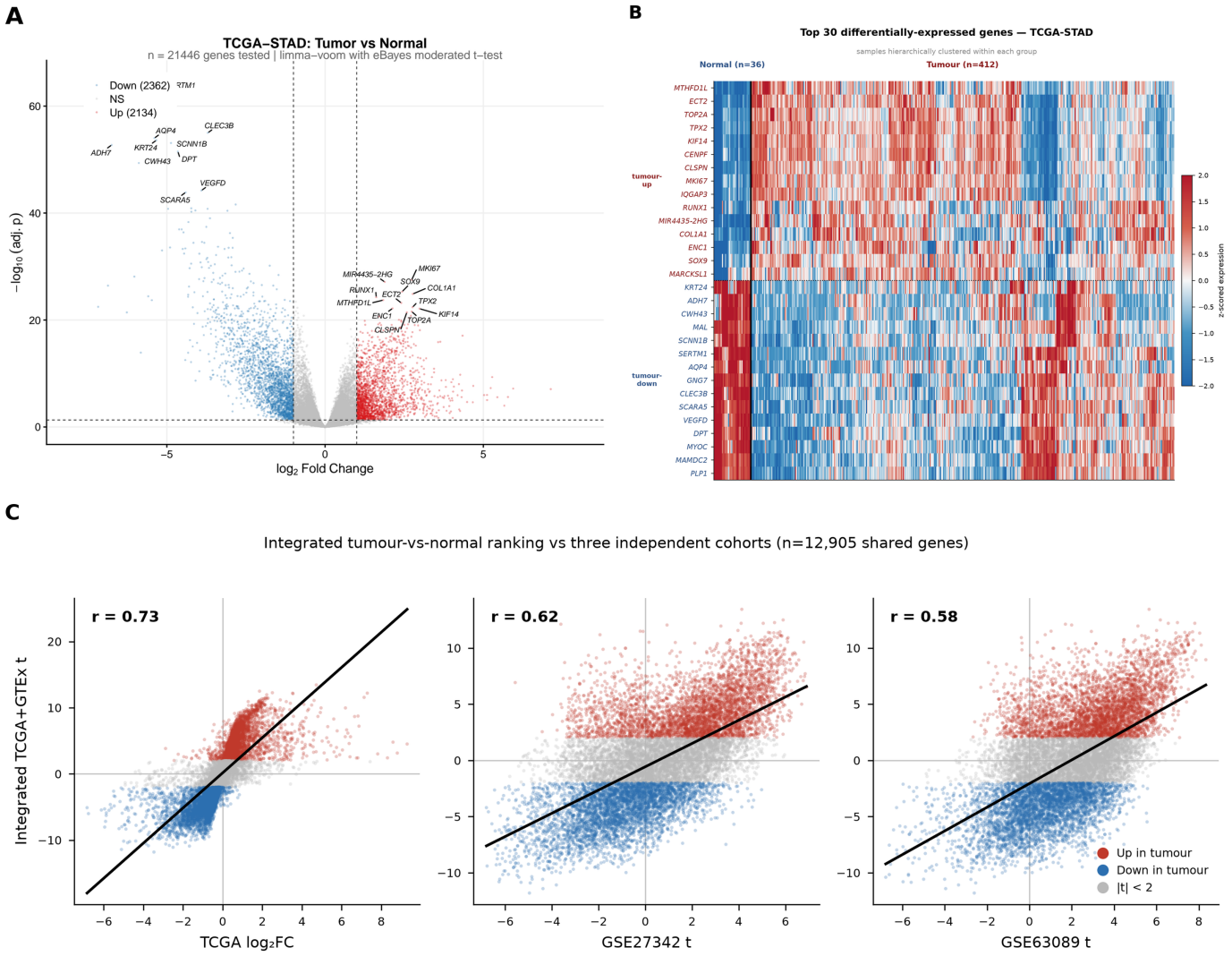


Figure 6. TCGA-only differential expression.
results/figures/Fig6.png

Figure 7

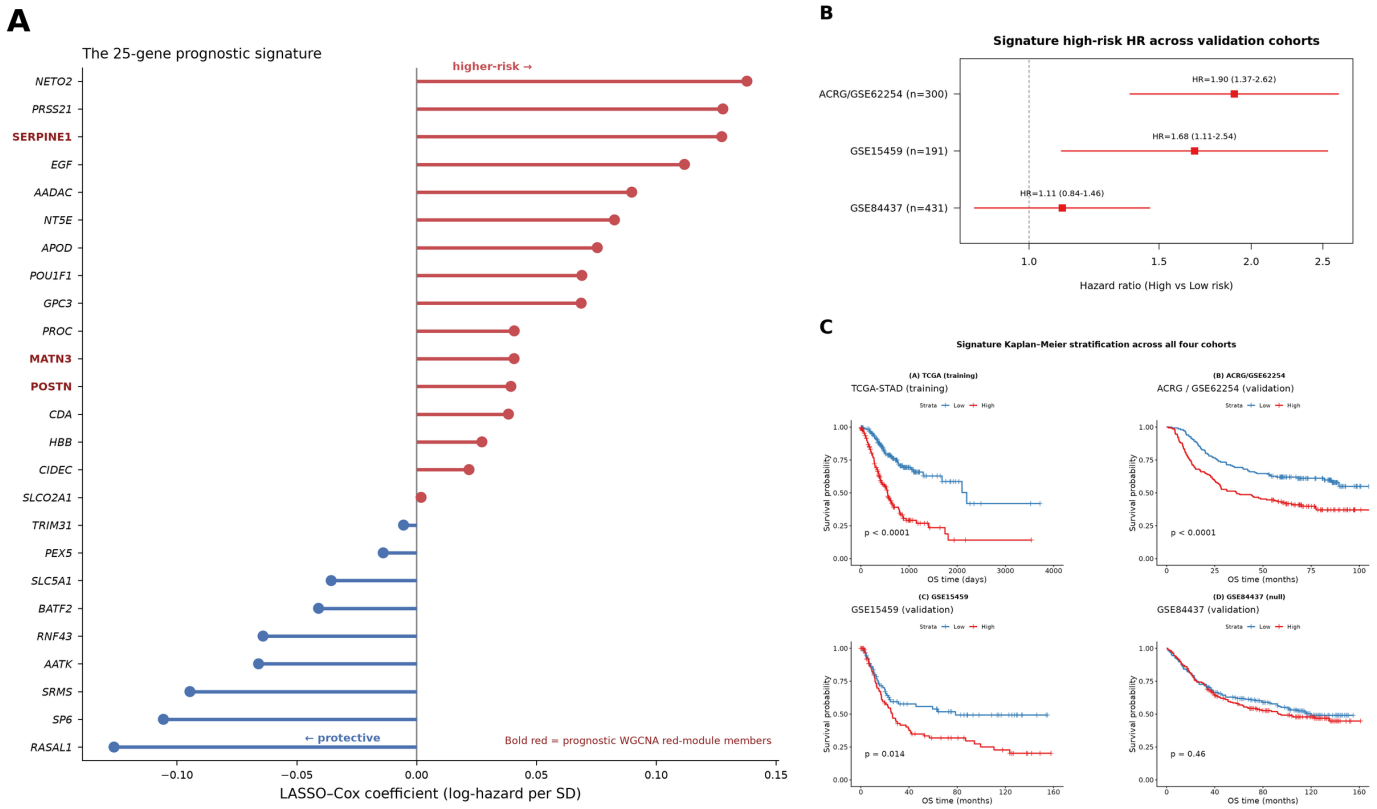


Figure 7. The 25-gene prognostic signature in detail.
results/figures/Fig7.png

Figure 8

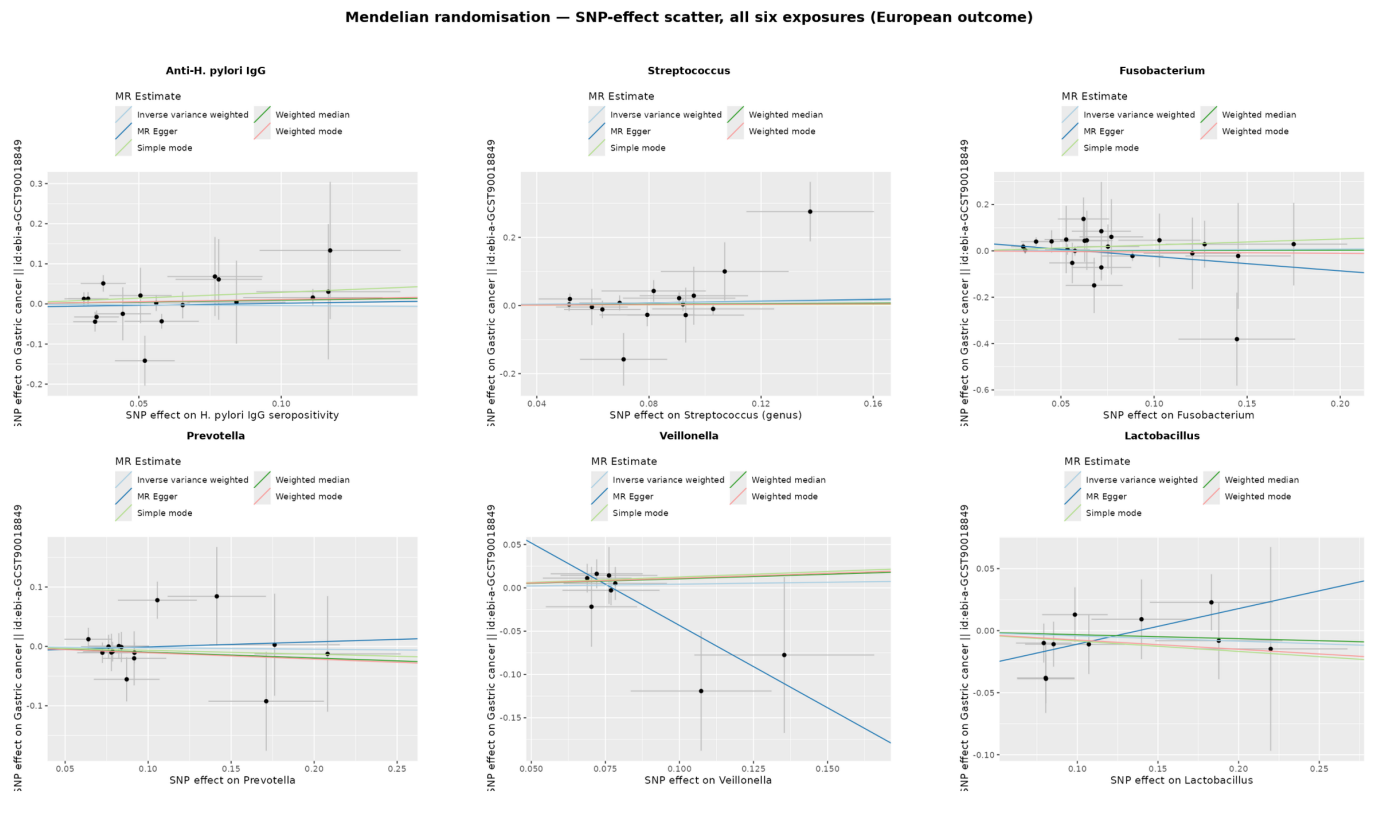
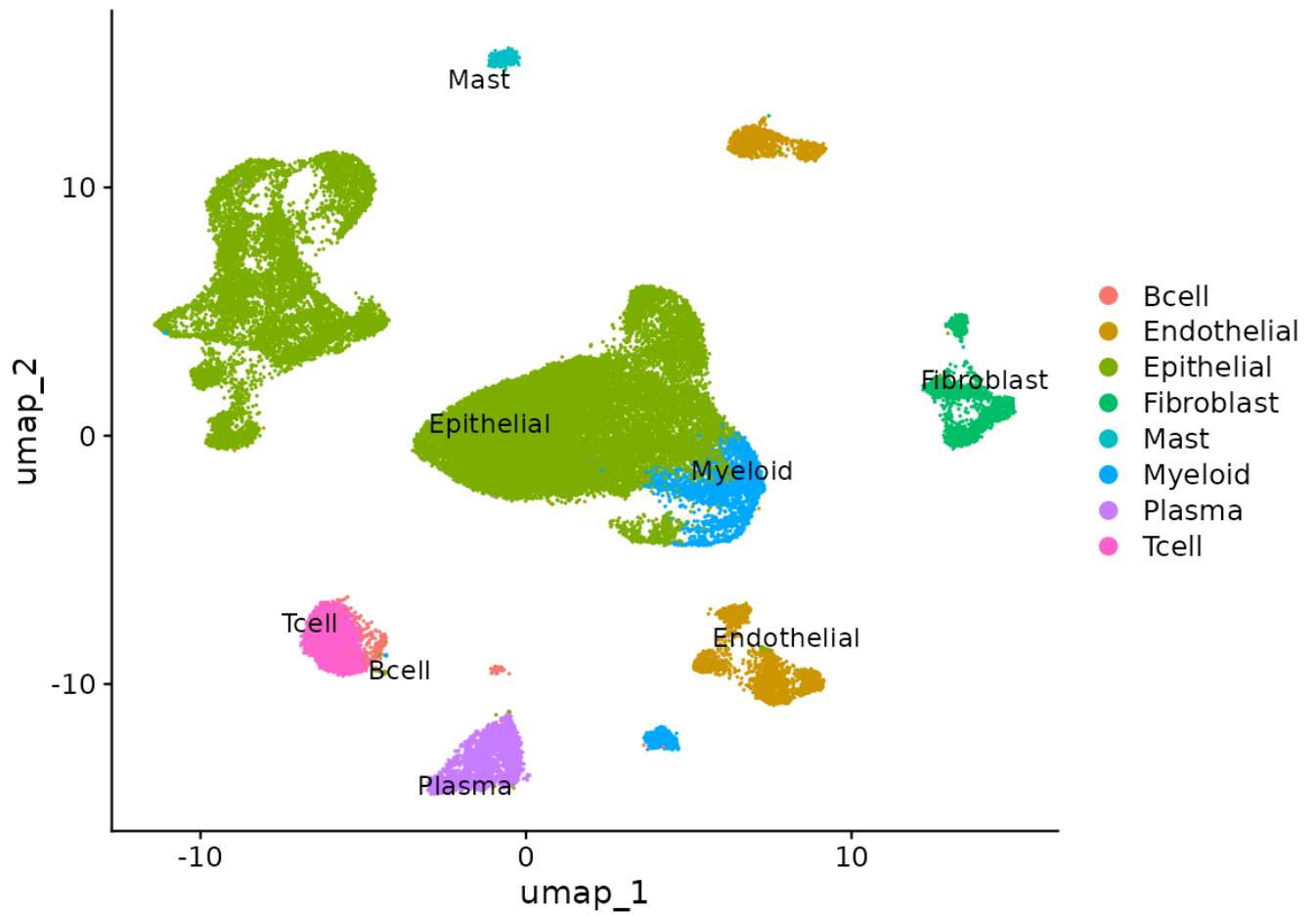


Figure 8. Mendelian-randomisation SNP-effect scatter, all six exposures.
results/figures/Fig8.png

Supplementary Figure S1

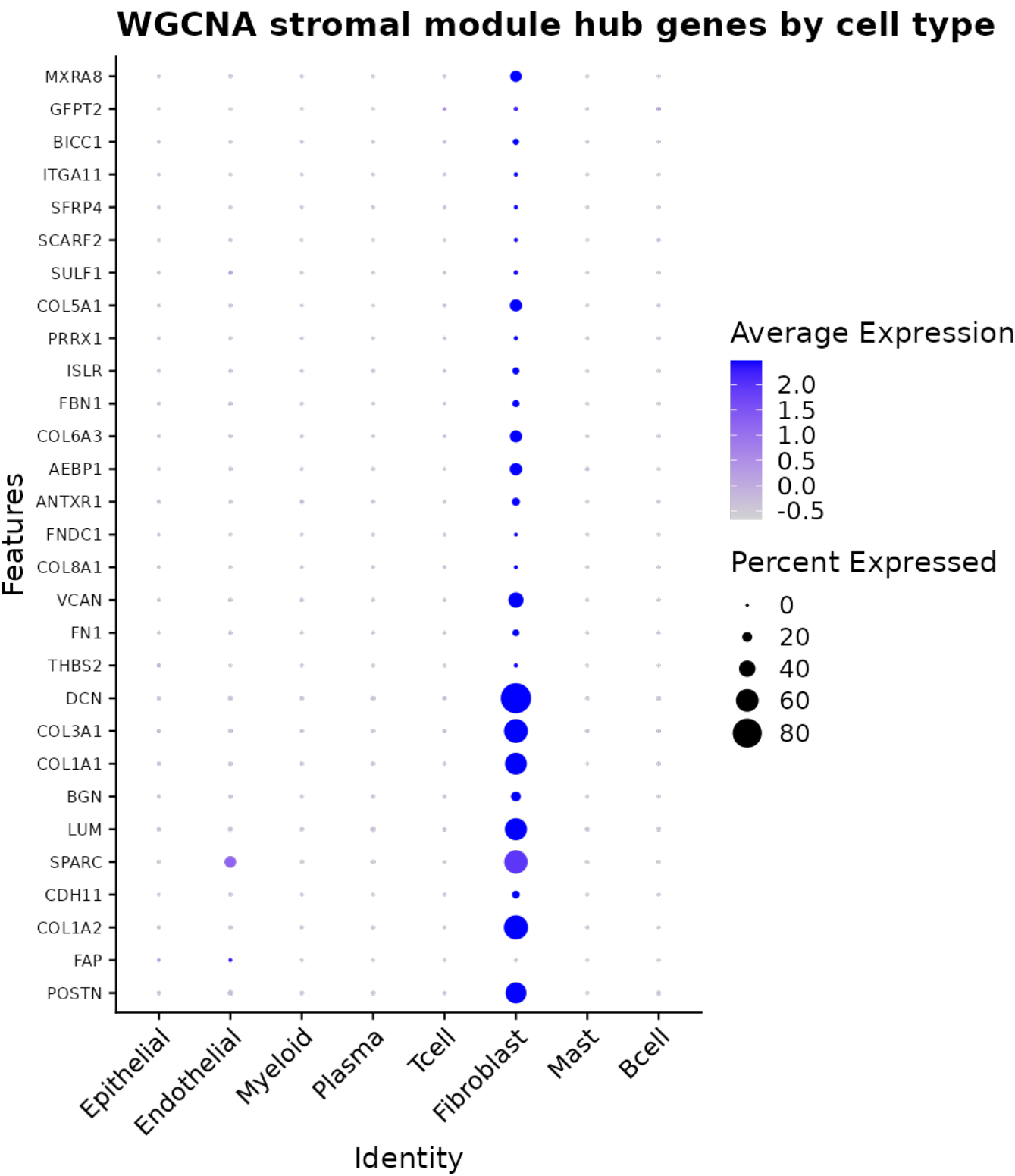
GSE134520 gastric mucosa scRNA-seq — cell types



Supplementary Figure S1. Single-cell atlas (GSE134520).

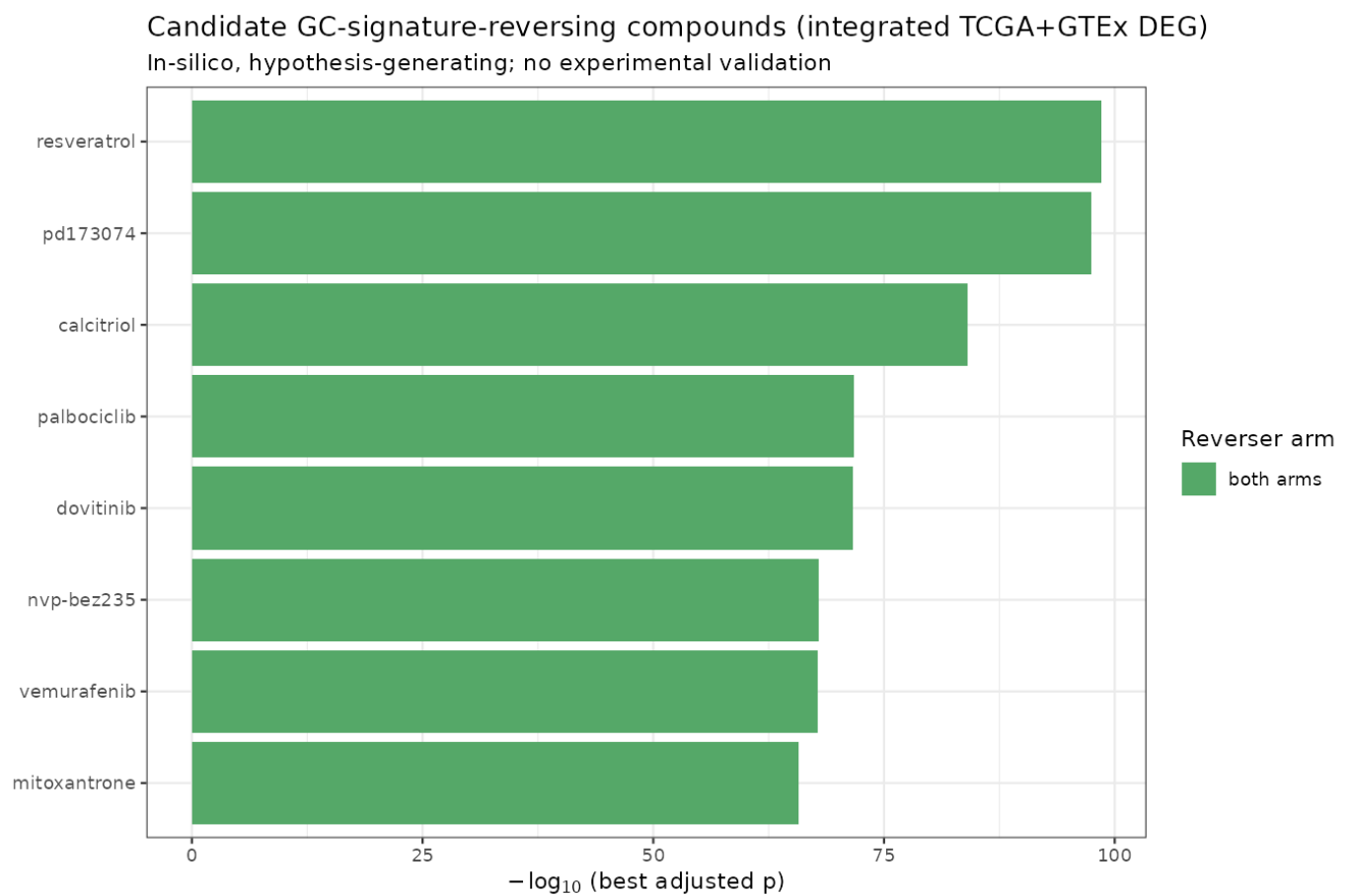
results/scrna/UMAP_celltypes.png

Supplementary Figure S2



Supplementary Figure S2. Stromal hub-gene expression by cell type.
results/scrna/DotPlot_stromal_module_hub.png

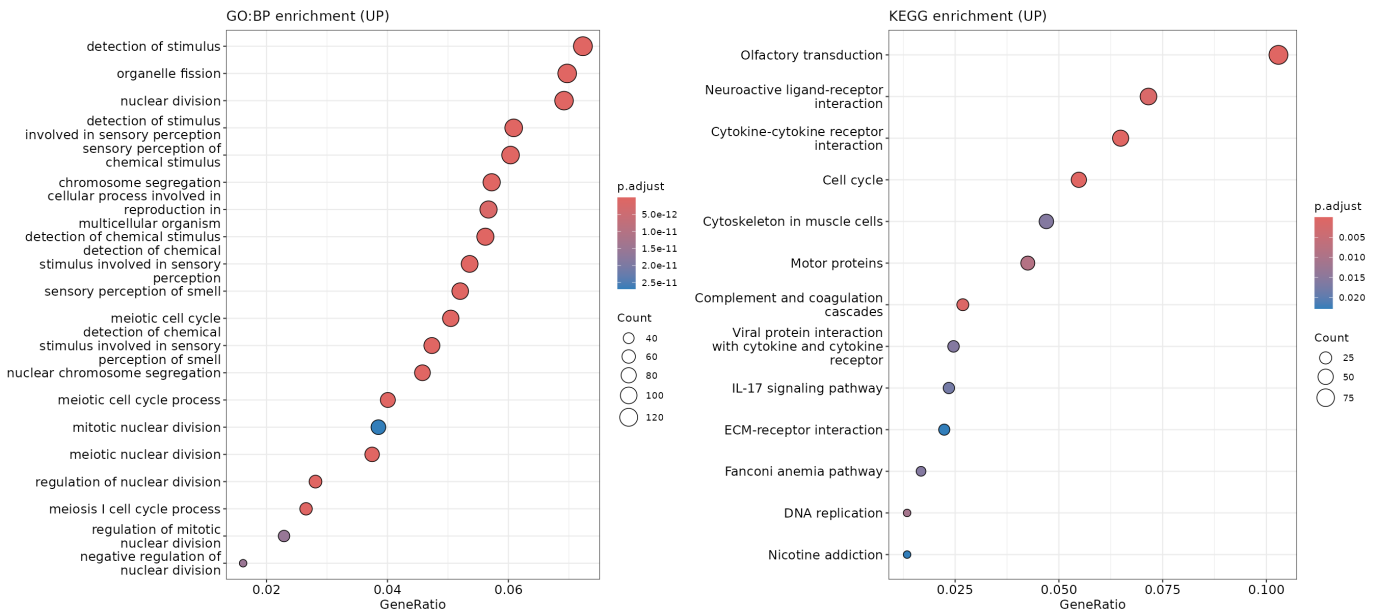
Supplementary Figure S3



Supplementary Figure S3. In-silico drug-repurposing candidates.

results/drug_repurposing_integrated/top_candidate_drugs.png

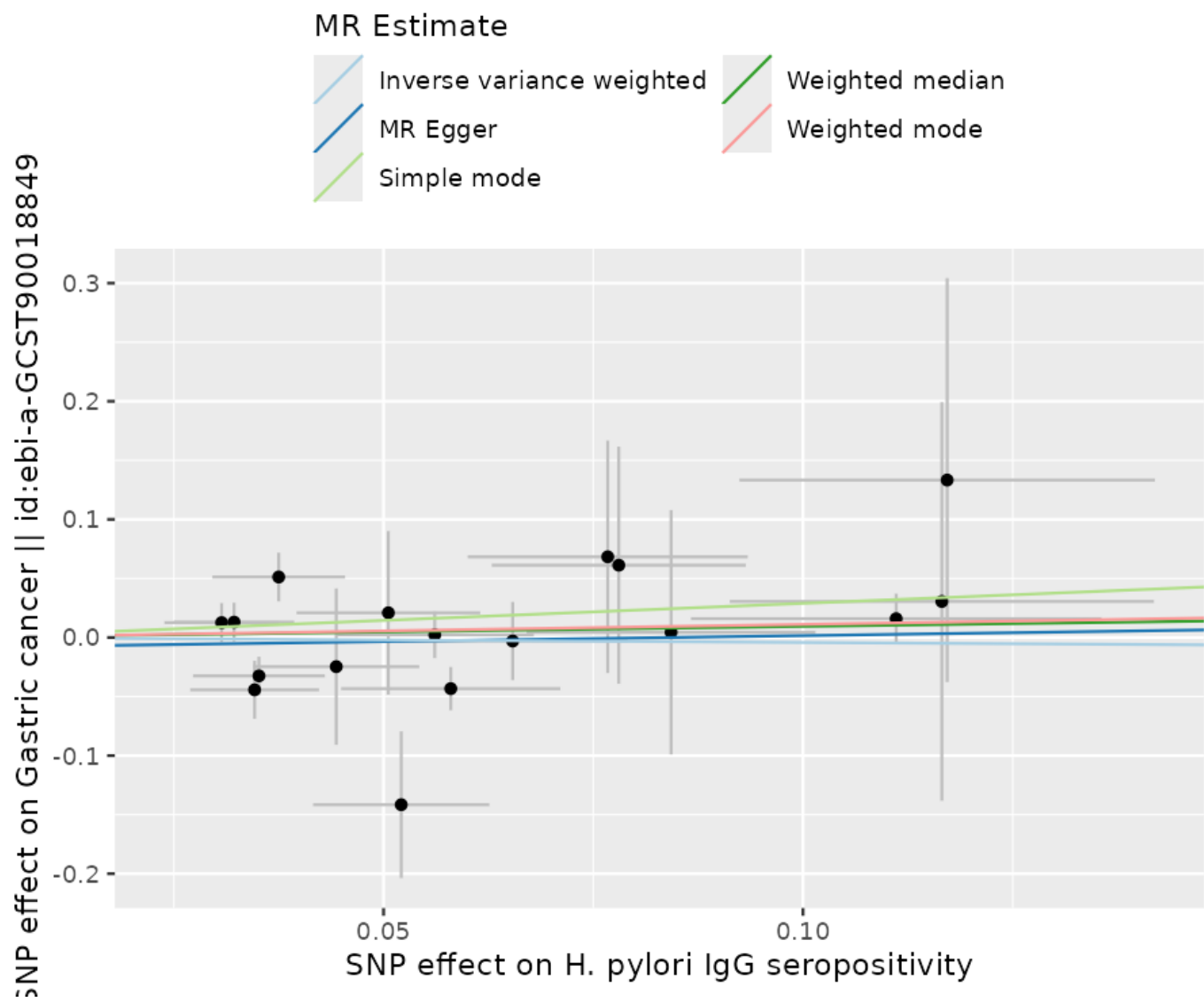
Supplementary Figure S4



Supplementary Figure S4. Over-representation analysis (GO:BP and KEGG, tumour-up genes).

results/enrichment/path_ORA_GO_KEGG.png

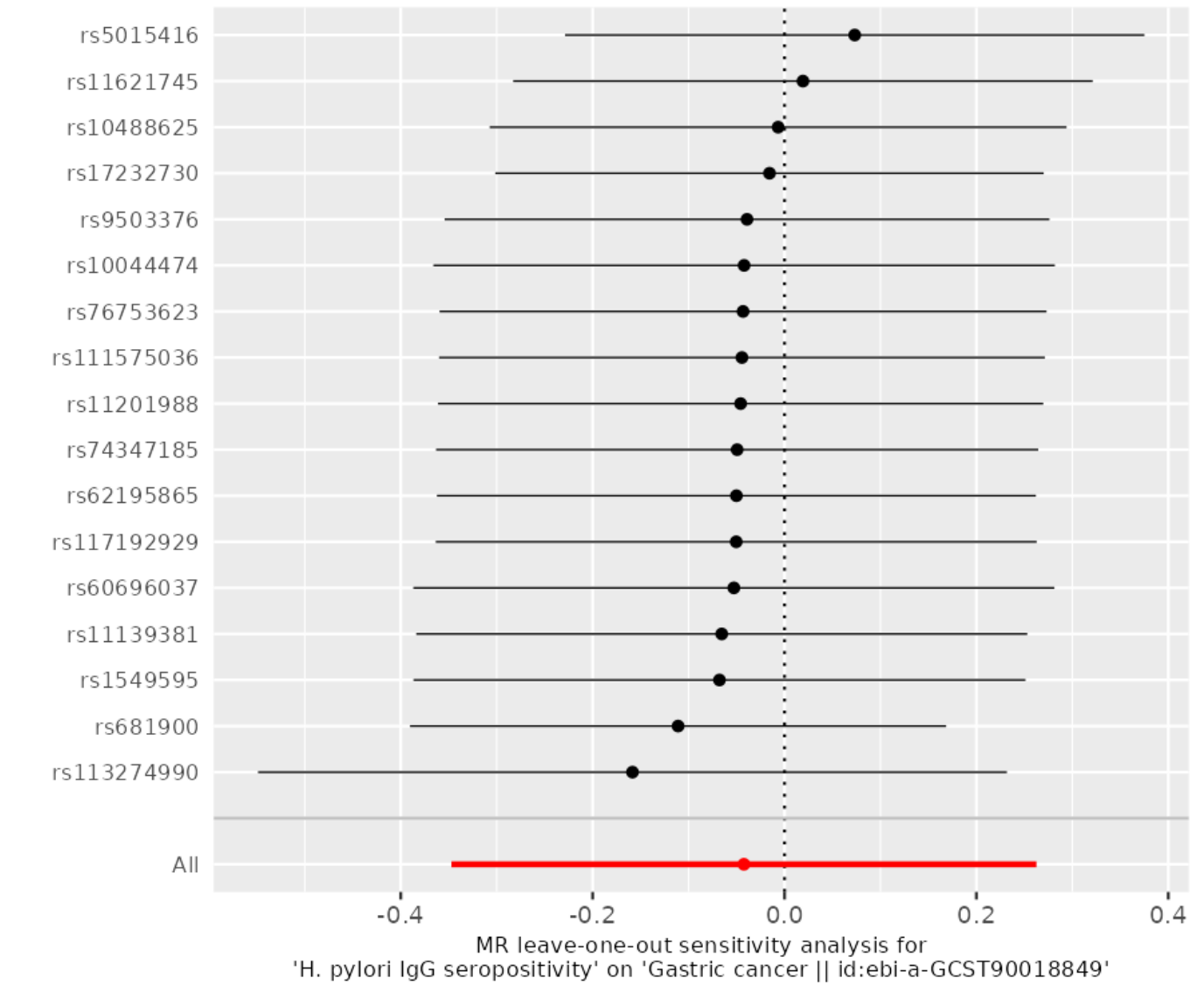
Supplementary Figure S5



Supplementary Figure S5. Mendelian-randomisation scatter (representative exposure).

results/mr_real/scatter_H_pylori_IgG_seropositivity.png

Supplementary Figure S6

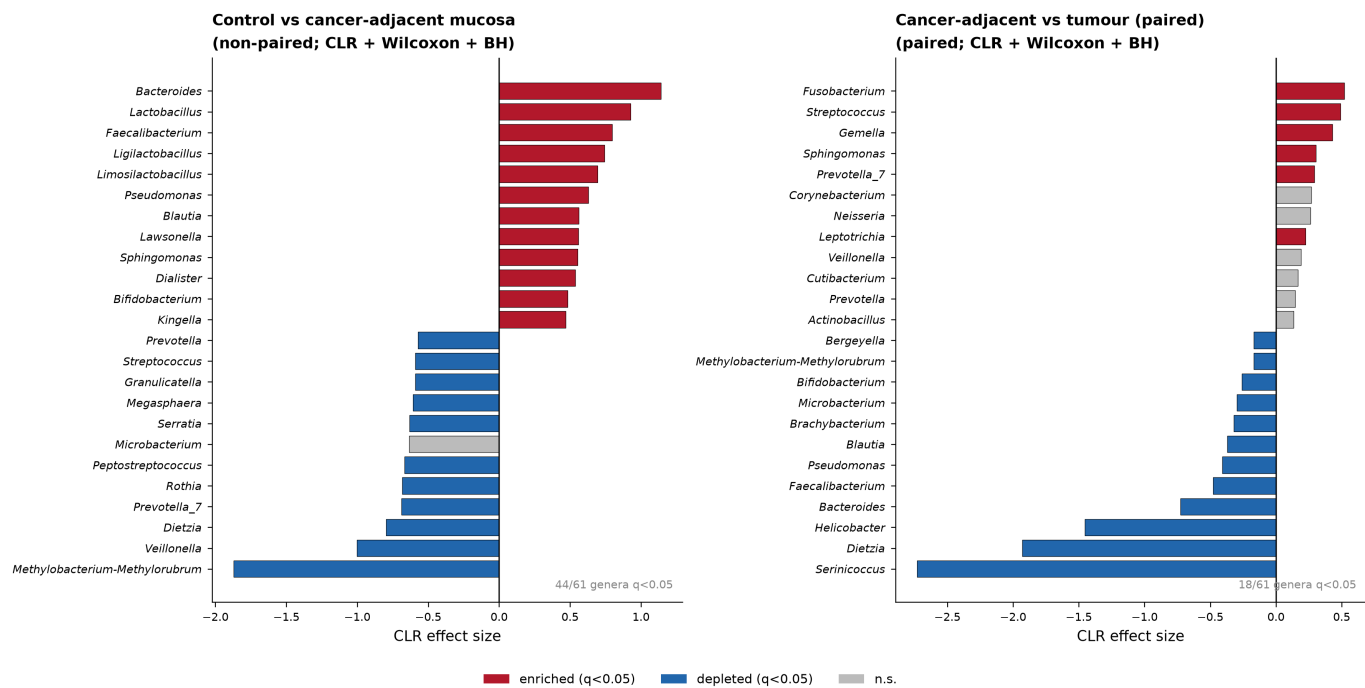


Supplementary Figure S6. Mendelian-randomisation leave-one-out (representative exposure).

results/mr_real/loo_H_pylori_IgG_seropositivity.png

Supplementary Figure S7

Tissue-microbiome differential abundance (compositional; batch-cautioned)

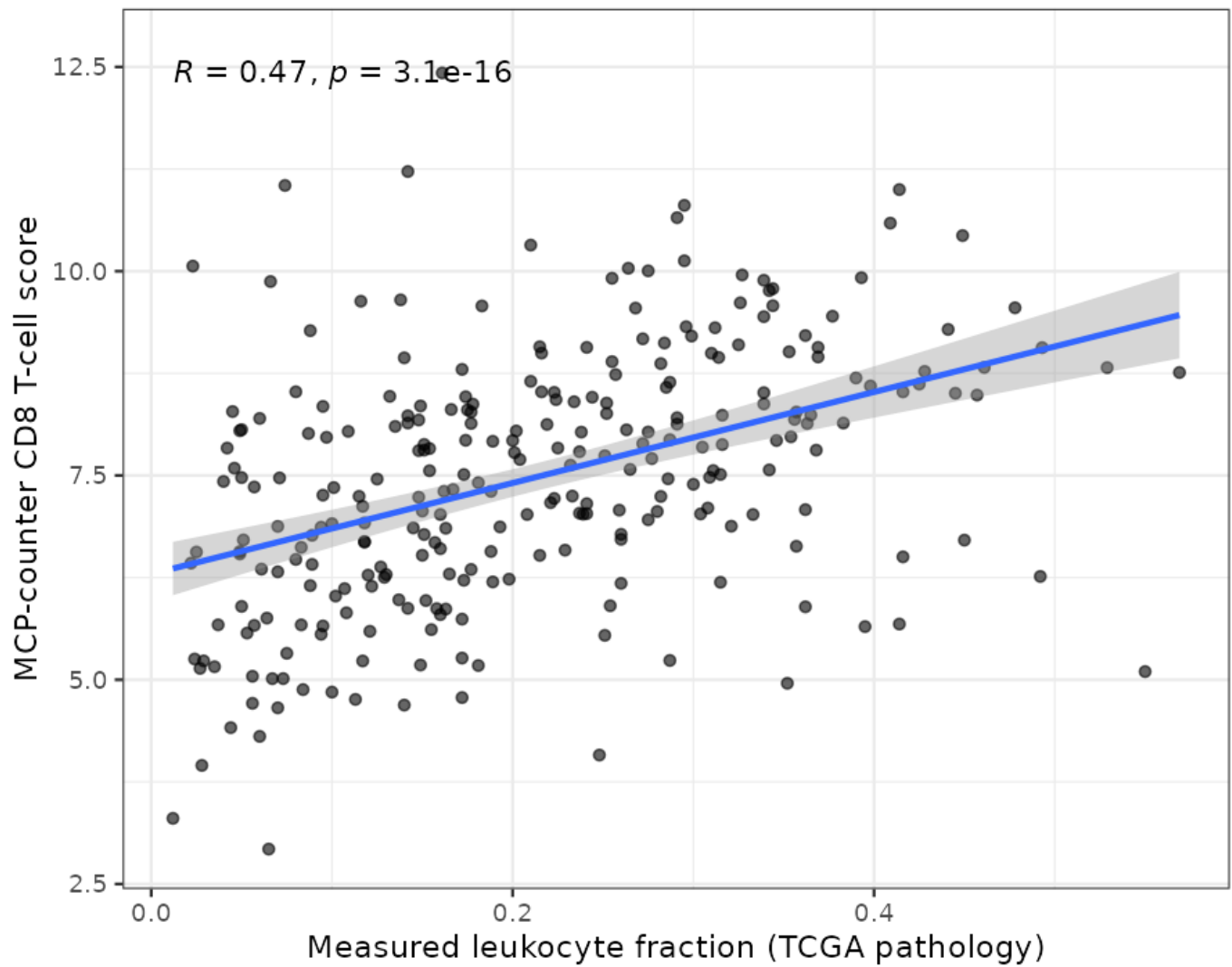


Supplementary Figure S7. Tissue-microbiome compositional differential abundance.

results/microbiome_biomarker/da_clr_barplot.png

Supplementary Figure S8A

MCP-counter CD8 T cells vs measured leukocyte % (tumors

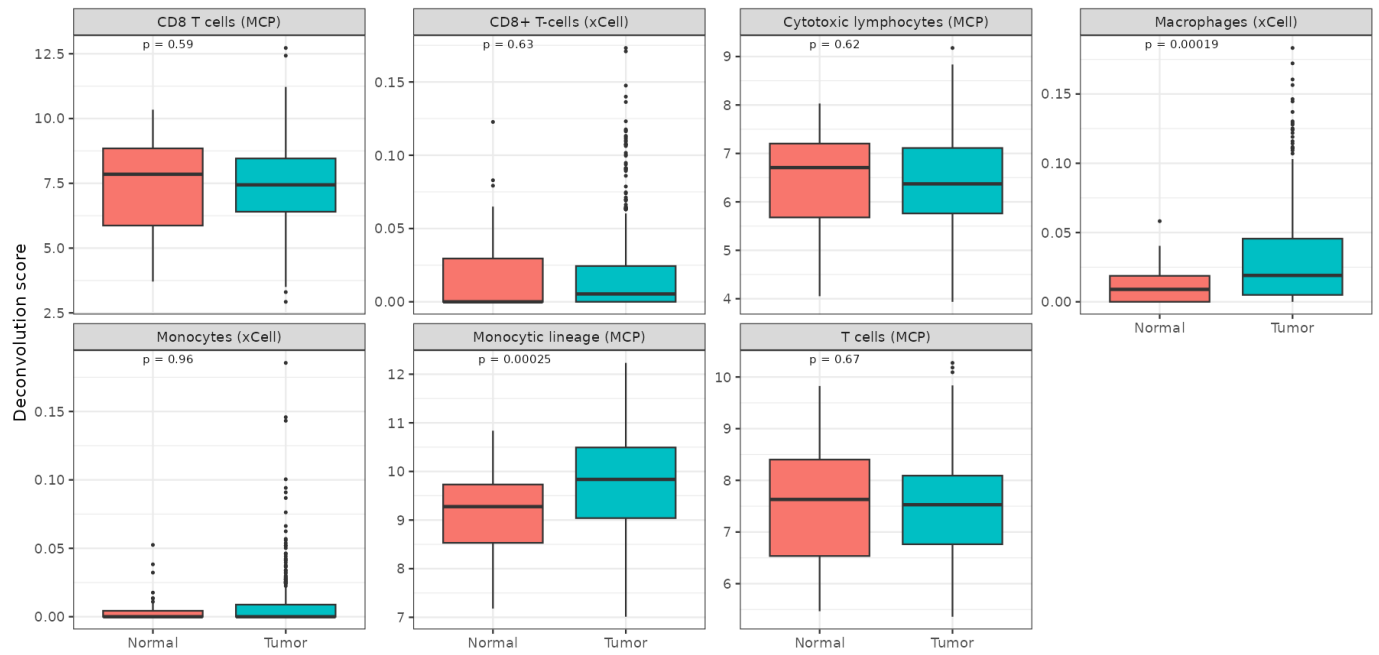


Deconvolution validated against measured leukocyte percentage (Spearman rho = 0.666, $p = 3.6e-36$, $n = 272$).

results/plots/Immune_validation_scatter.png

Supplementary Figure S8B

Immune populations: Tumor vs Normal (TCGA-STAD)

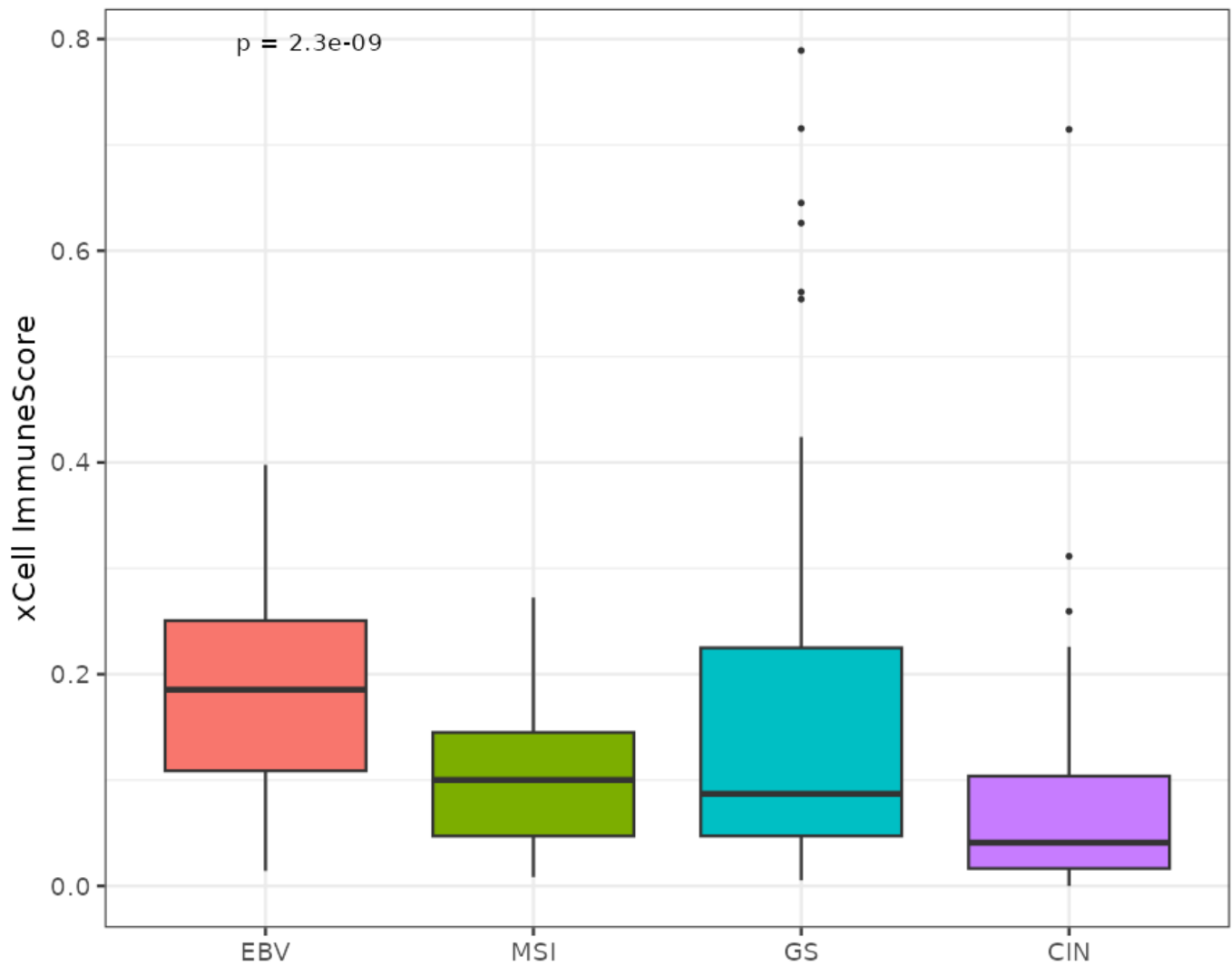


Immune cell-type estimates, tumour versus normal tissue.

results/plots/Immune_tumor_vs_normal.png

Supplementary Figure S8C

xCell ImmuneScore by molecular subtype (TCGA-STAD)

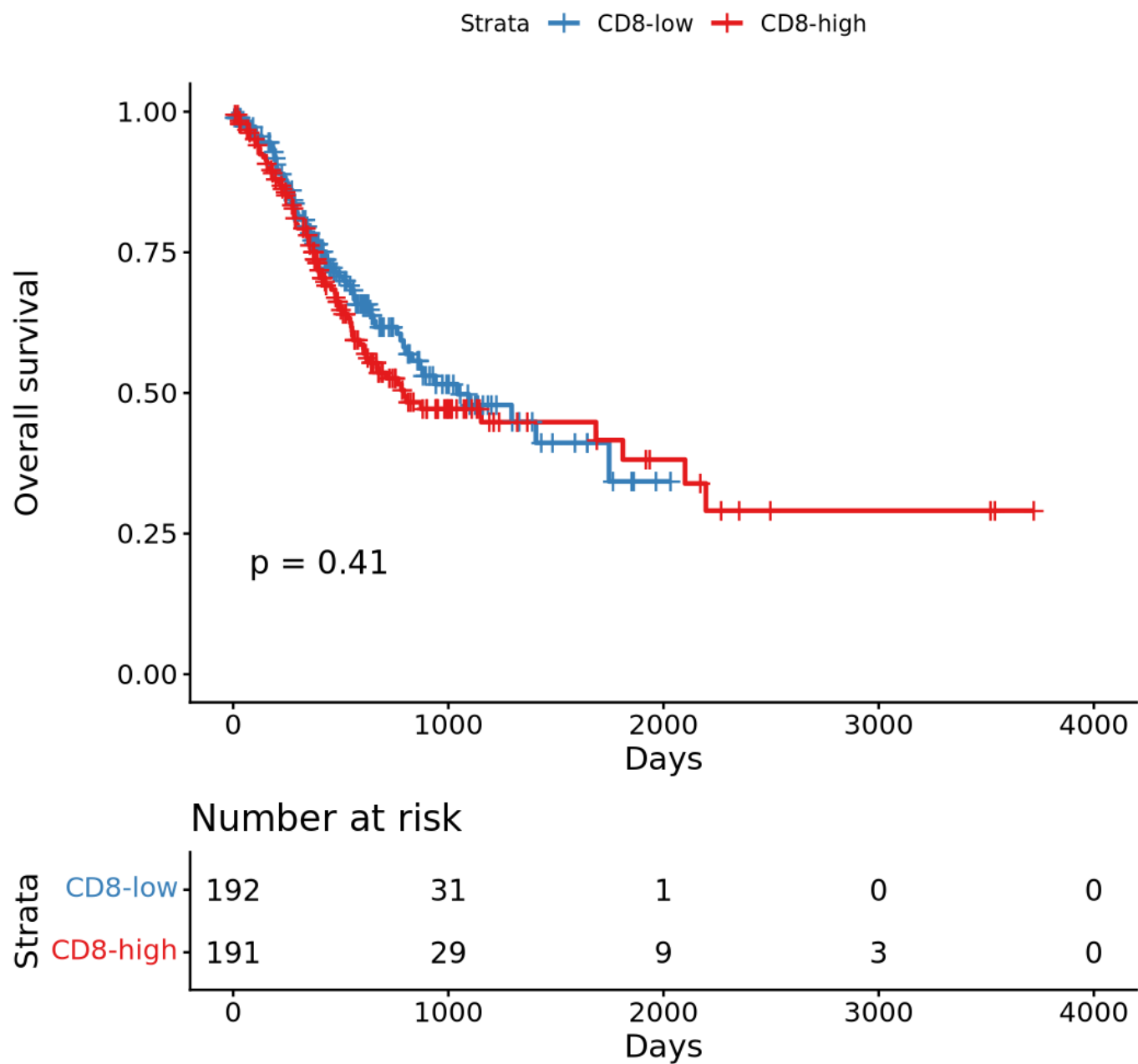


Immune infiltration by Lauren subtype.

results/plots/Immune_by_subtype.png

Supplementary Figure S8D

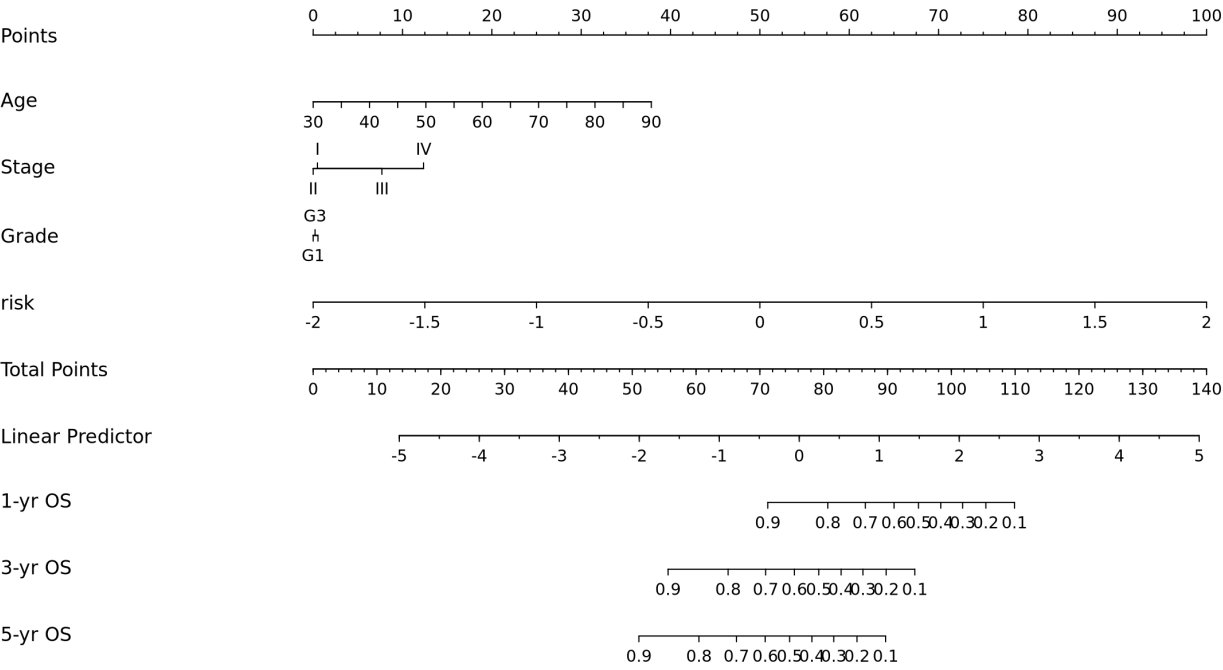
OS by MCP-counter CD8 T-cell score (median split)



CD8 T-cell infiltration and overall survival (HR 1.04, $p = 0.41$; not prognostic).

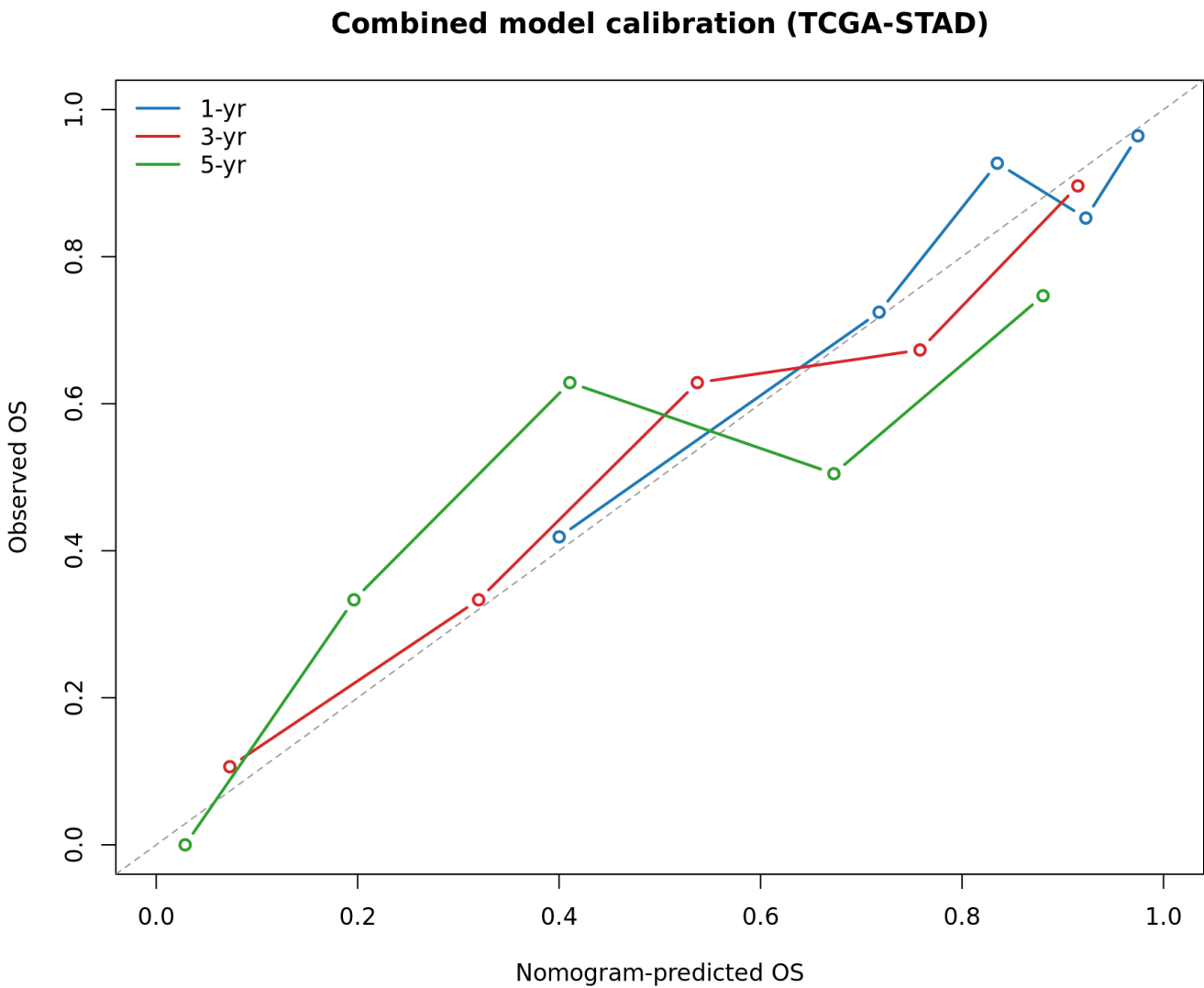
results/plots/Immune_CD8_survival_KM.png

Supplementary Figure S9A



Clinical nomogram combining stage, age and the 25-gene signature.

[results/nomogram_combined/combined_nomogram.png](#)

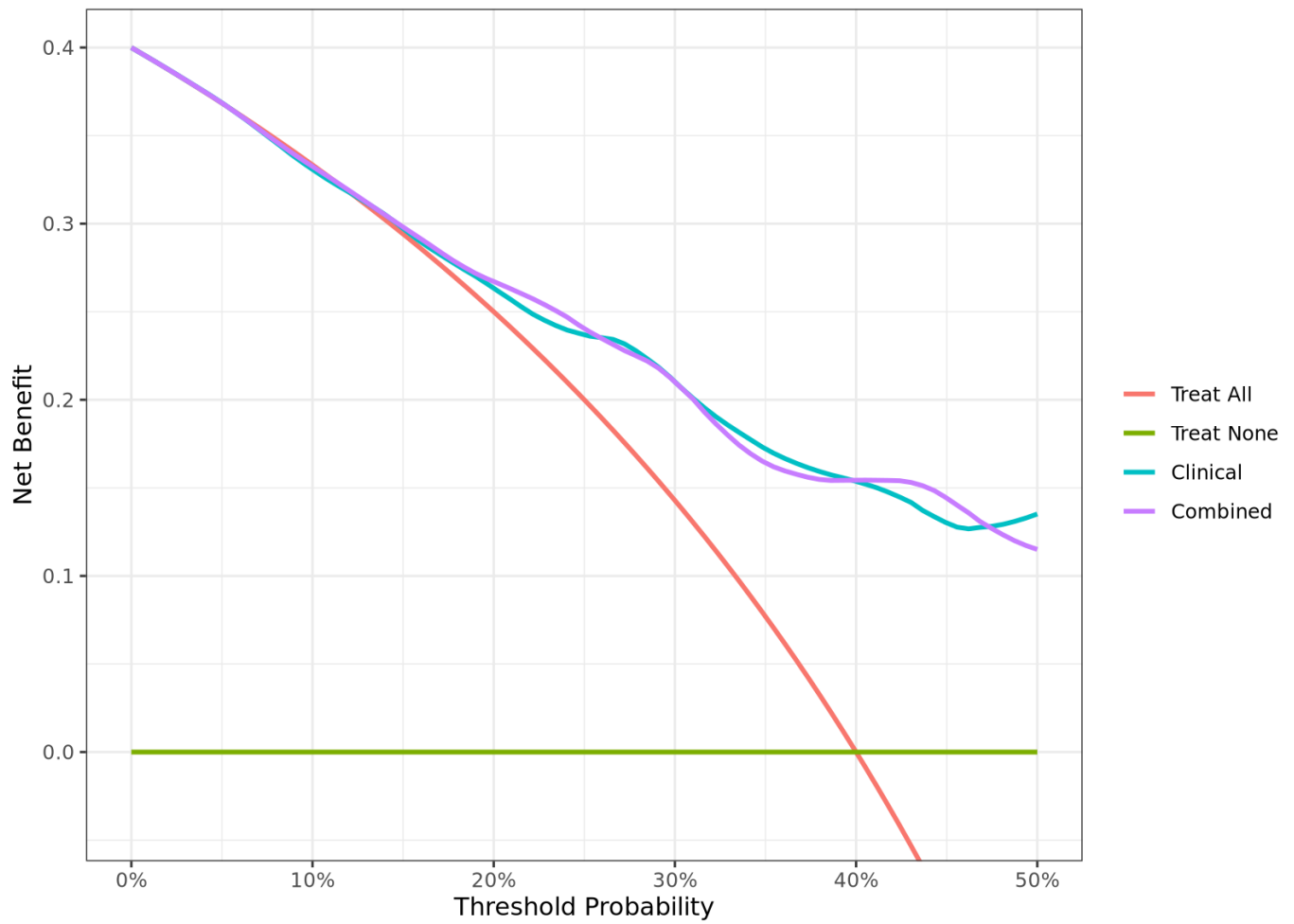


Calibration of the combined model against observed survival.

results/nomogram_combined/calibration_combined.png

Supplementary Figure S9C

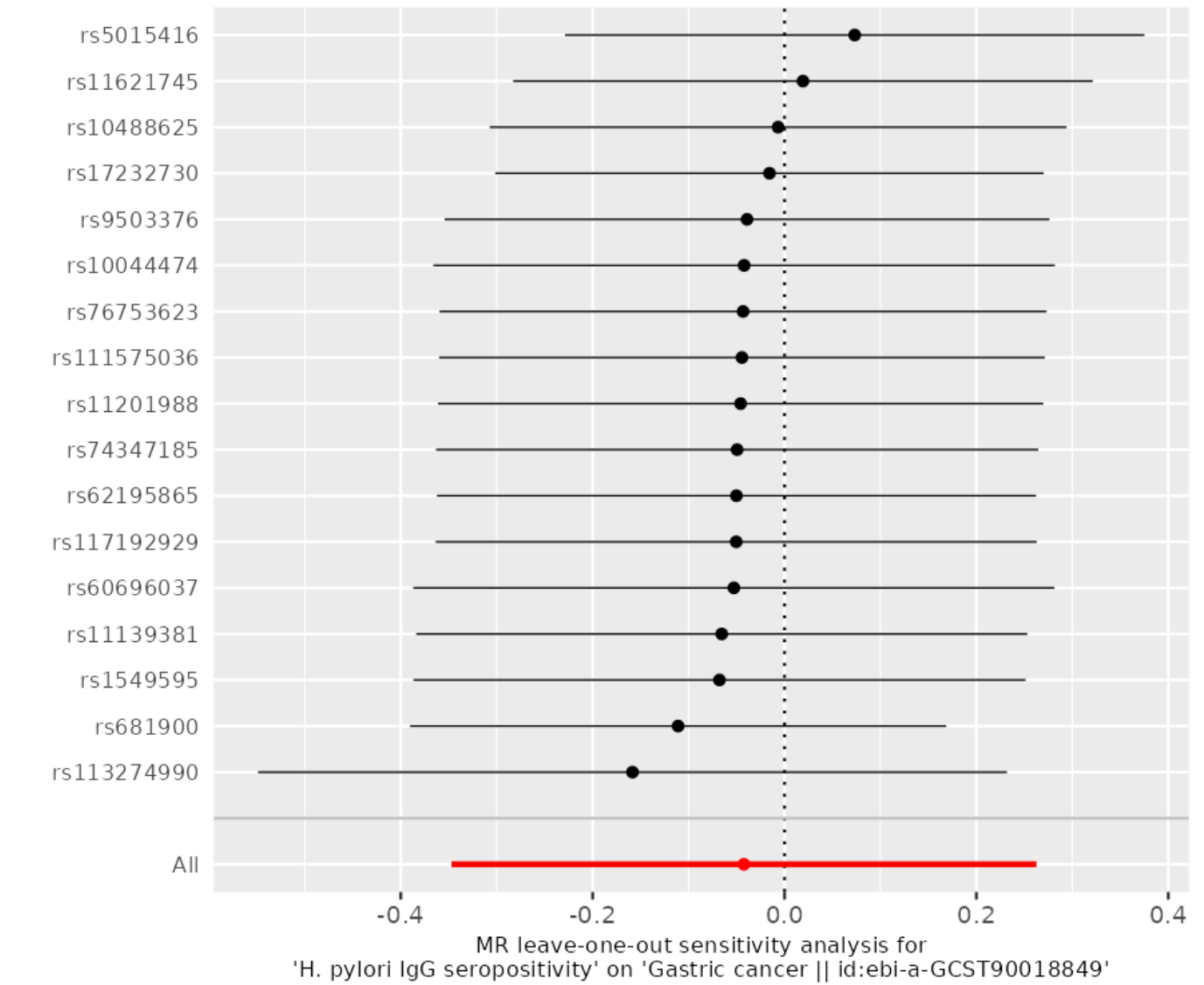
External DCA @3y (ACRG/GSE62254)



External decision-curve analysis in ACRG: no incremental net benefit over staging (delta-C +0.005).

results/external_utility_ACRG/DCA_external.png

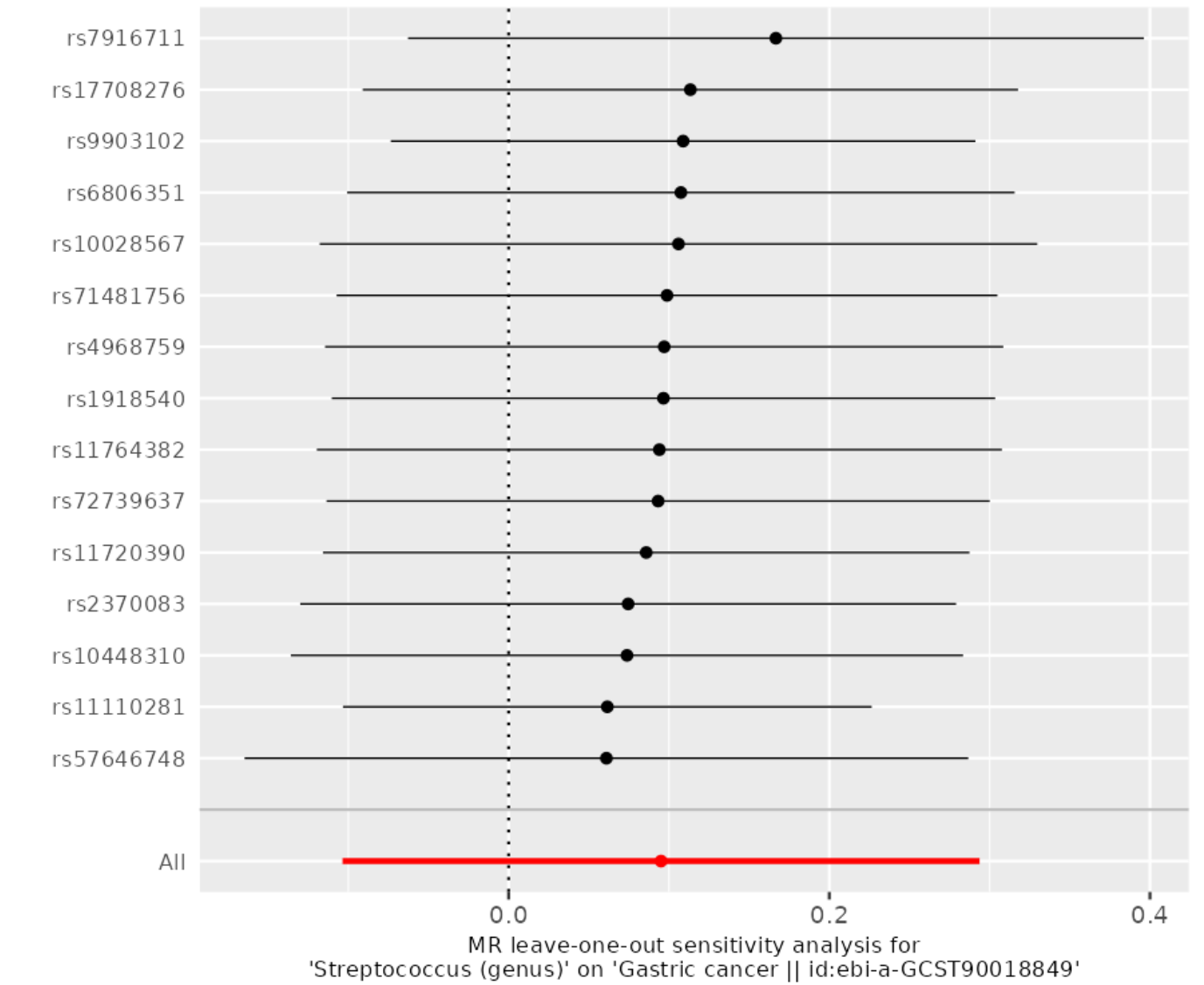
Supplementary Figure S10A



Mendelian-randomisation leave-one-out: anti-H. pylori IgG seropositivity.

results/mr_real/loo_H_pylori_IgG_seropositivity.png

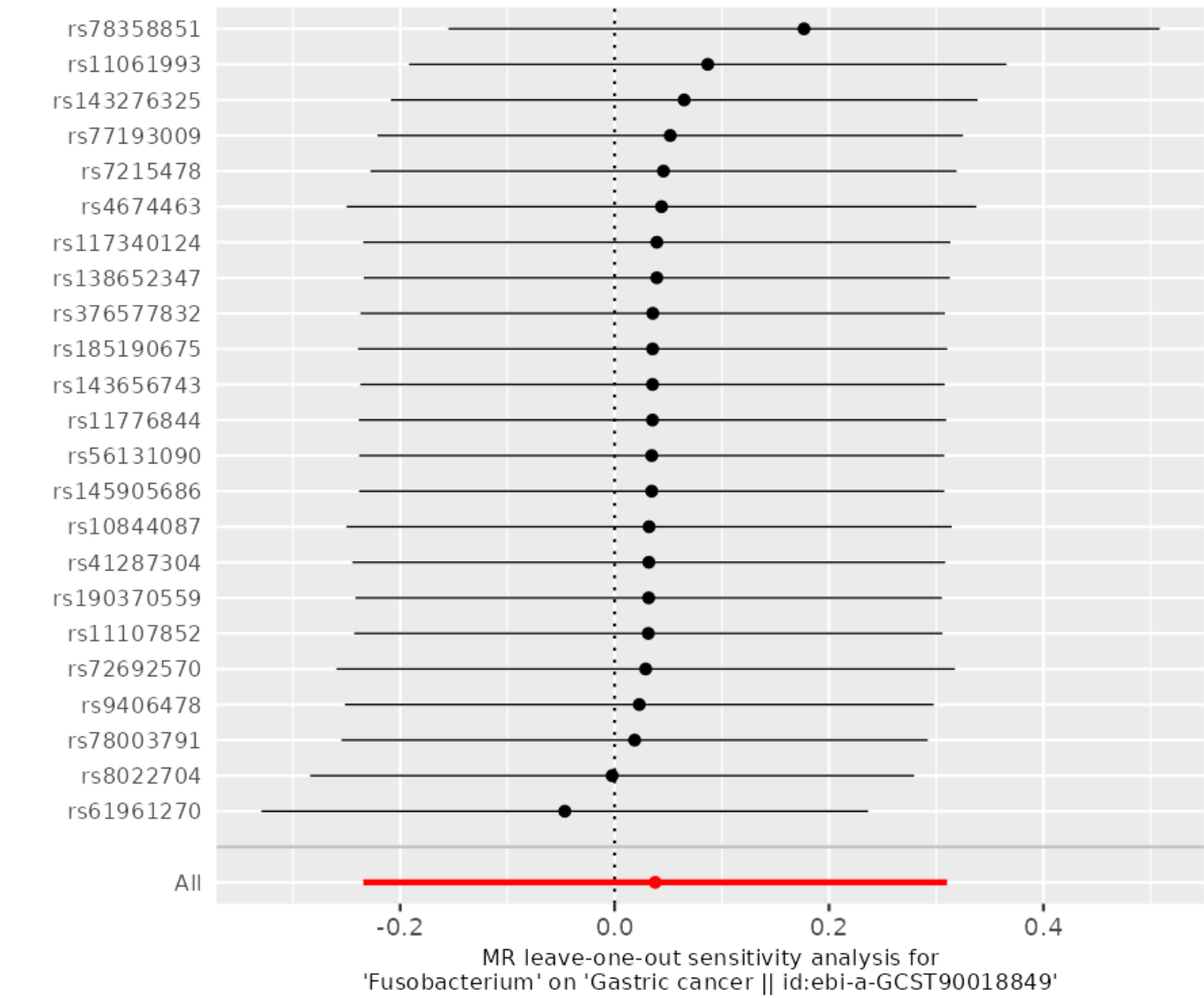
Supplementary Figure S10B



Mendelian-randomisation leave-one-out: Streptococcus (genus).

results/mr_real/loo_Streptococcus__genus_.png

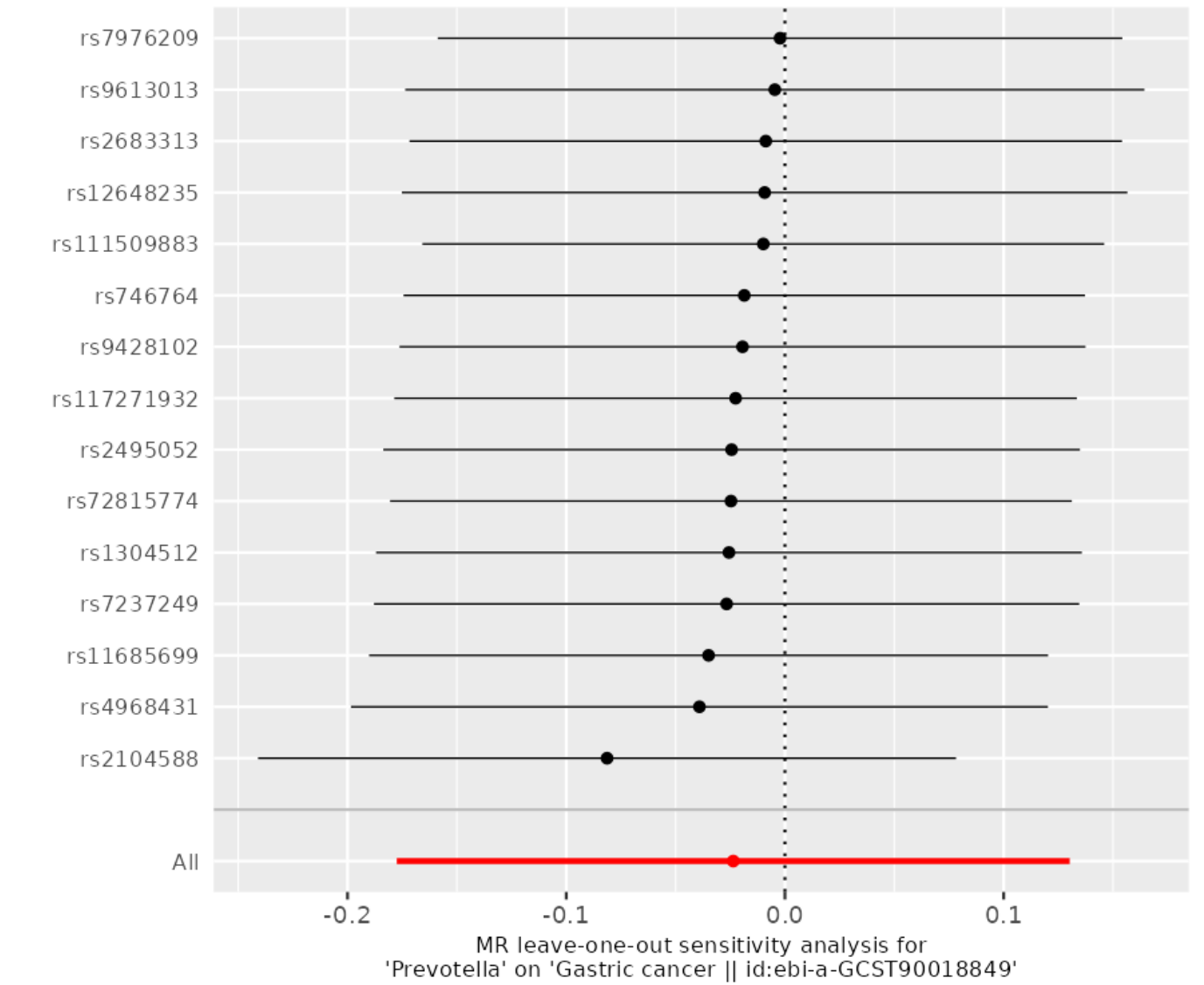
Supplementary Figure S10C



Mendelian-randomisation leave-one-out: Fusobacterium.

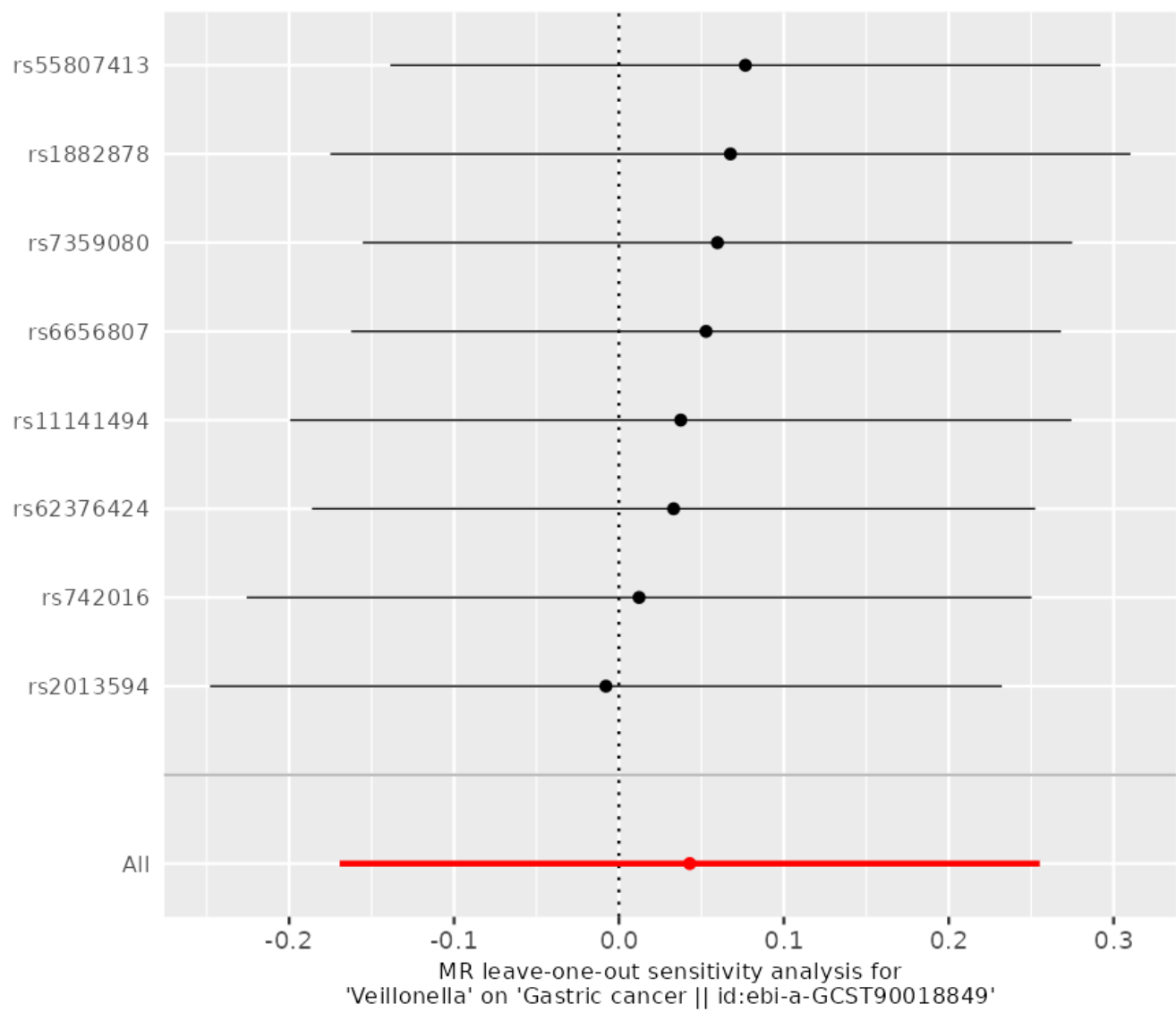
results/mr_real/loo_Fusobacterium.png

Supplementary Figure S10D



Mendelian-randomisation leave-one-out: Prevotella.
results/mr_real/loo_Prevotella.png

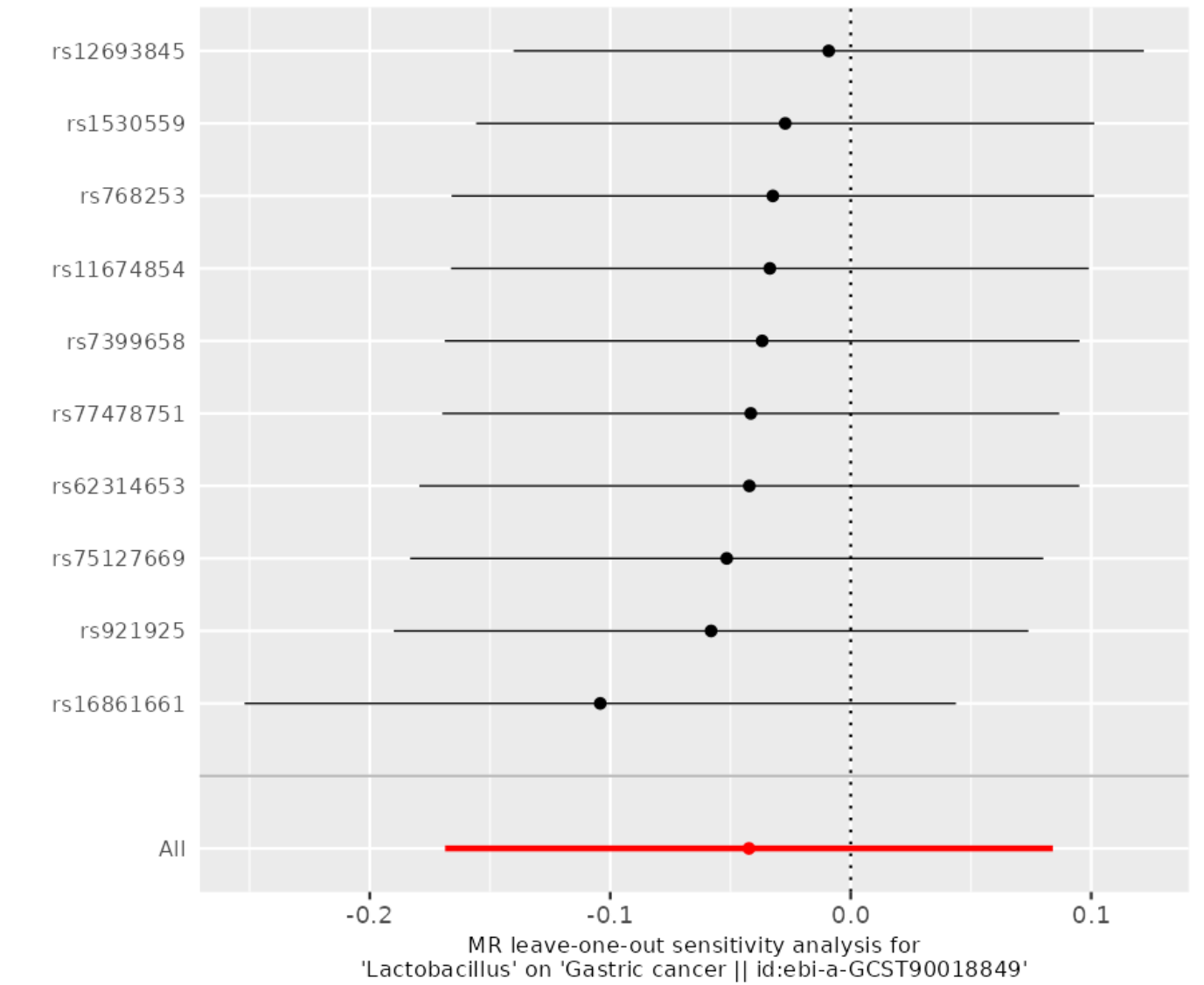
Supplementary Figure S10E



Mendelian-randomisation leave-one-out: Veillonella.

results/mr_real/loo_Veillonella.png

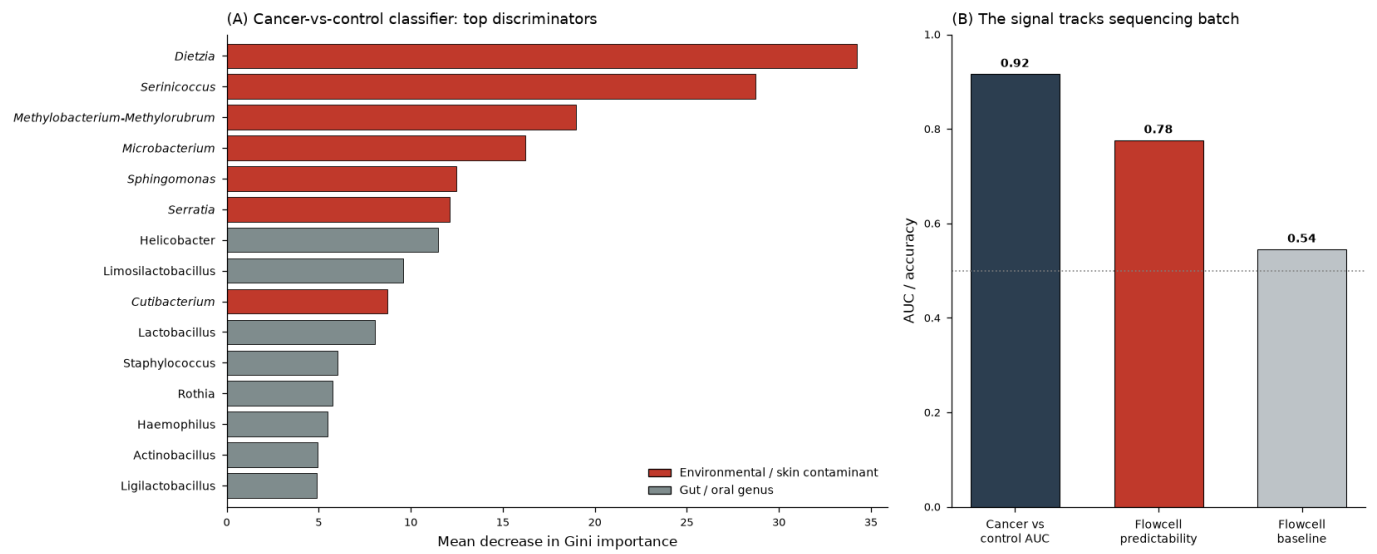
Supplementary Figure S10F



Mendelian-randomisation leave-one-out: Lactobacillus.
results/mr_real/loo_Lactobacillus.png

Supplementary Figure S11

Tumour-microbiome random forest: apparent accuracy is a batch artefact



Supplementary Figure S11. The tumour-microbiome classifier is a batch artefact.

results/microbiome_biomarker/rf_batch_classifier.png